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Semester:

4th Semester

Subject:

Database Applications

Lab Task:

SQL Server – DDL Commands

Submitted To:

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Lab Task 6

Section 1:

Creating a Database and Table Using SQL DDL Commands

1. Create a new database named as your full name in all lowercase with underscores, followed by _lab_task_04.



New

- collegedb
- college_db
- information_schema
- muhammad_bilal_khan_lab_task_06
- mysql
- performance_schema
- phpmyadmin

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0031 seconds.)

`CREATE DATABASE muhammad_bilal_khan_lab_task_06;`

[Edit inline] [Edit] [Create PHP code]

⚠ Error: #1046 No database selected

2. In this database, create a table named Students with the following attributes and appropriate data types:

StudentID (INT, Primary Key)

FirstName (VARCHAR, 50)

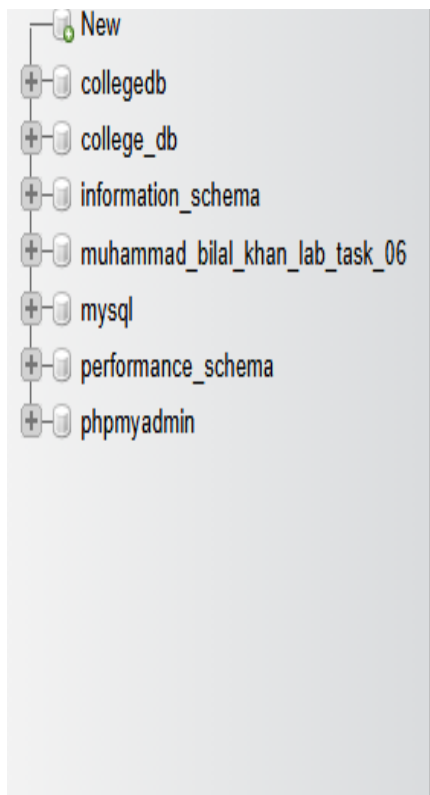
LastName (VARCHAR, 50)

DOB (DATE)

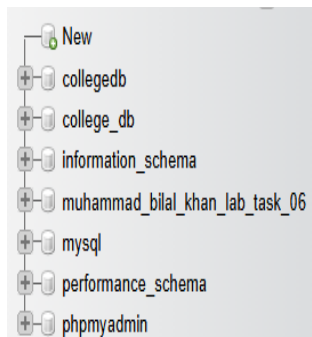
Gender (VARCHAR, 10)

Email (VARCHAR, 100)

PhoneNumber (VARCHAR, 15)



```
1 CREATE TABLE Students
2 (
3 StudentID INT AUTO_INCREMENT PRIMARY KEY,
4 FirstName VARCHAR(50) NOT NULL,
5 LastName VARCHAR(50) NOT NULL,
6 DOB DATE,
7 GENDER VARCHAR(10) NOT NULL,
8 Email VARCHAR(100) NOT NULL,
9 PhoneNumber VARCHAR(15)
10 );
```



✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

```
USE muhammad_bilal_khan_lab_task_06;
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

⚠ Error: #1046 No database selected

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0176 seconds.)

```
CREATE TABLE Students ( StudentID INT AUTO_INCREMENT PRIMARY KEY, FirstName VARCHAR(50) NOT NULL, LastName VARCHAR(50) NOT NULL, DOB
DATE, GENDER VARCHAR(10) NOT NULL, Email VARCHAR(100) NOT NULL, PhoneNumber VARCHAR(15) );
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

New

- collegedb
- college_db
- information_schema
- muhammad_bilal_khan_lab_task_06
 - New
 - students
- mysql
- performance_schema
- phpmyadmin

```
SELECT * FROM `students`
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

StudentID	FirstName	LastName	DOB	GENDER	Email	PhoneNumber
-----------	-----------	----------	-----	--------	-------	-------------

Query results operations

Create view

Section 2:

Modifying Table Structure Using SQL DDL Commands

1. Add a new column Address of type VARCHAR(255) to the Students table.

New

- collegedb
- college_db
- information_schema
- muhammad_bilal_khan_lab_task_06
 - New
 - students
- mysql
- performance_schema
- phpmyadmin

```
1 ALTER TABLE students ADD Address VARCHAR(255);
```

Clear Format Get auto-saved query

Navigation tree:

- New
 - collegedb
 - New
 - students
 - college_db
 - information_schema
 - muhammad_bilal_khan_lab_task_06
 - New
 - students
 - mysql
 - performance_schema
 - phpmyadmin

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0228 seconds.)

```
ALTER TABLE students ADD Address VARCHAR(255);
```

[Edit inline] [Edit] [Create PHP code]

Navigation tree:

- New
 - collegedb
 - New
 - students
 - college_db
 - information_schema
 - muhammad_bilal_khan_lab_task_06
 - New
 - students
 - mysql
 - performance_schema
 - phpmyadmin

```
SELECT * FROM `students`
```

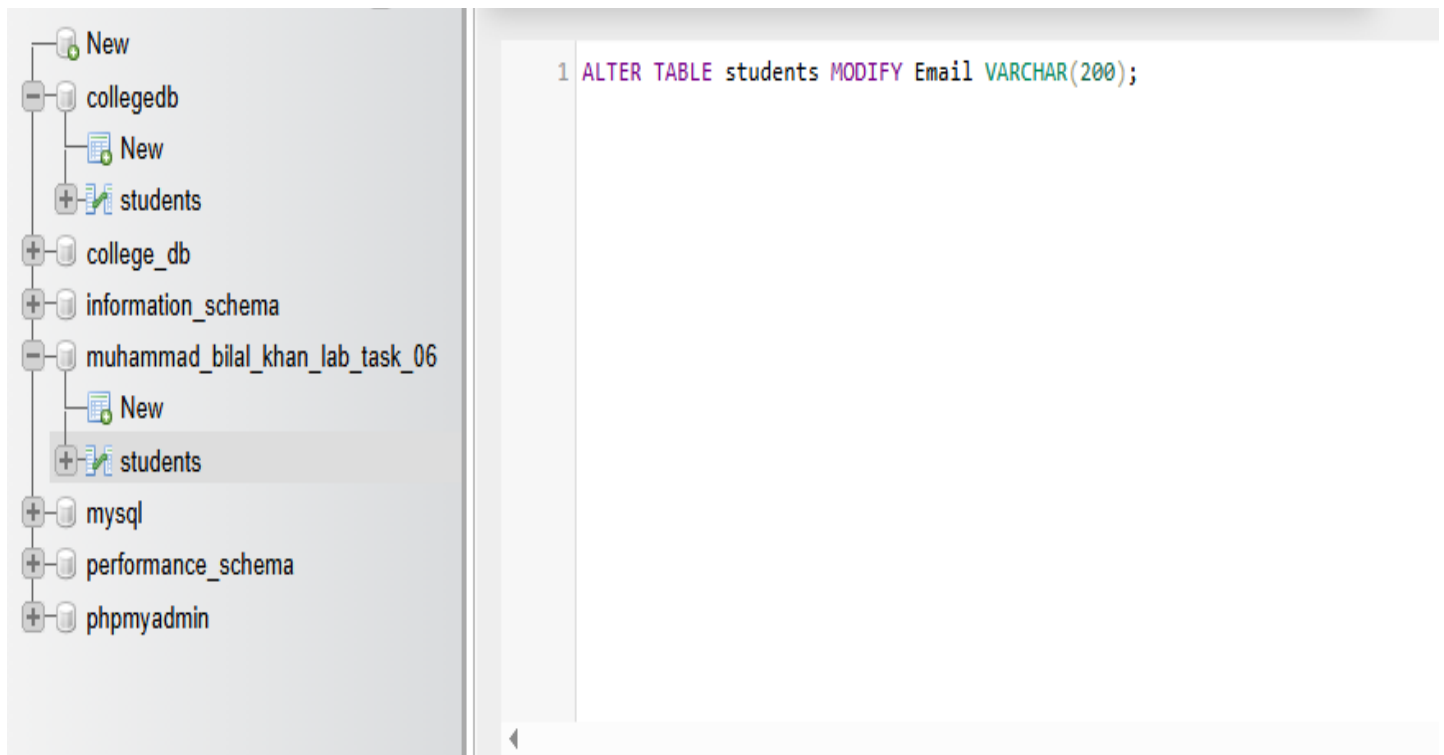
☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

StudentID	FirstName	LastName	DOB	GENDER	Email	PhoneNumber	Address
-----------	-----------	----------	-----	--------	-------	-------------	---------

Query results operations

Create view

2. Modify the column Email to be of type VARCHAR(200).



The screenshot shows a database management interface. On the left, a tree view displays the database structure, with the 'students' table under the 'collegedb' database selected. On the right, the SQL query editor contains the following command:

```
1 ALTER TABLE students MODIFY Email VARCHAR(200);
```



The screenshot shows the same database management interface after the query has been executed. A green message bar at the top right indicates the successful execution of the query:

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0542 seconds.)

Below the message bar, the executed SQL query is displayed:

```
ALTER TABLE students MODIFY Email VARCHAR(200);
```

At the bottom of the query editor, there are links for further actions: [\[Edit inline \]](#), [\[Edit \]](#), and [\[Create PHP code \]](#).

Recent
Favorites

New
collegedb
college_db
information_schema
muhammad_bilal_khan_lab_task_06
New
students
mysql
performance_schema
phpmyadmin

Table structure
Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	StudentID	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	FirstName	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐ 3	LastName	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐ 4	DOB	date			Yes	NULL			Change Drop More
☐ 5	GENDER	varchar(10)	utf8mb4_general_ci		No	None			Change Drop More
☐ 6	Email	varchar(200)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 7	PhoneNumber	varchar(15)	utf8mb4_general_ci		Yes	NULL			Change Drop More

☐ Check all
With selected:
 Browse
 Change
 Drop
 Primary
 Unique
 Index
 Spatial

 Fulltext

3. Remove the column PhoneNumber from the Students table

New
college_db
information_schema
muhammad_bilal_khan_lab_task
New
coursedetails
mysql
performance_schema
phpmyadmin

1 ALTER TABLE students DROP PhoneNumber;

The screenshot shows a database management interface. On the left is a tree view of the database structure:

- college_db
 - information_schema
 - muhammad_bilal_khan_lab_task_06
 - New
 - students
 - mysql
 - performance_schema
 - phpmyadmin

On the right, there is a toolbar with the following options: ☐ Profiling, [\[Edit inline \]](#), [\[Edit \]](#), [\[Explain SQL \]](#), [\[Create PHP code \]](#), and [\[Refresh \]](#).

Below the toolbar is a table header with the following columns: StudentID, FirstName, LastName, DOB, GENDER, Email, and Address.

Below the table header is a section titled "Query results operations" with a button labeled "Create view".

Section 3:

Create the Second Table

1. Create a new table named Courses with the following attributes:

CourseID (INT, Primary Key)

CourseName (VARCHAR, 100)

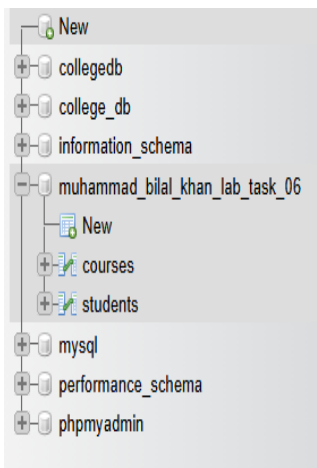
Instructor (VARCHAR, 50)

Credits (INT)

StartDate (DATE)

EndDate (DATE)

Department (VARCHAR, 50)



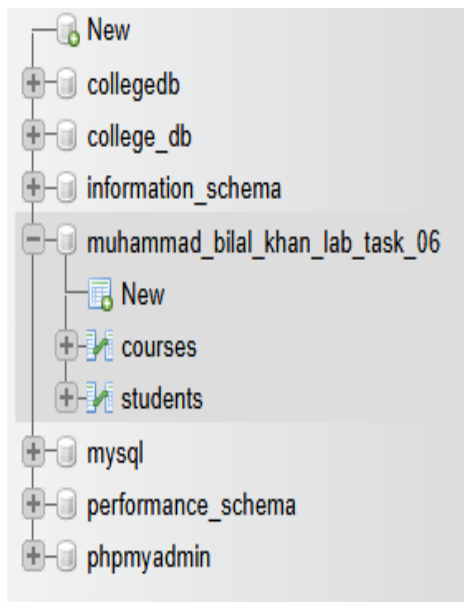
✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0337 seconds.)

```
CREATE TABLE Courses ( CourseID INT PRIMARY KEY, CourseName VARCHAR(100) NOT NULL, Instructor VARCHAR(50) NOT NULL, Credits INT, StartDate DATE, EndDate DATE, Department VARCHAR(50) NOT NULL );
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

Section 4:

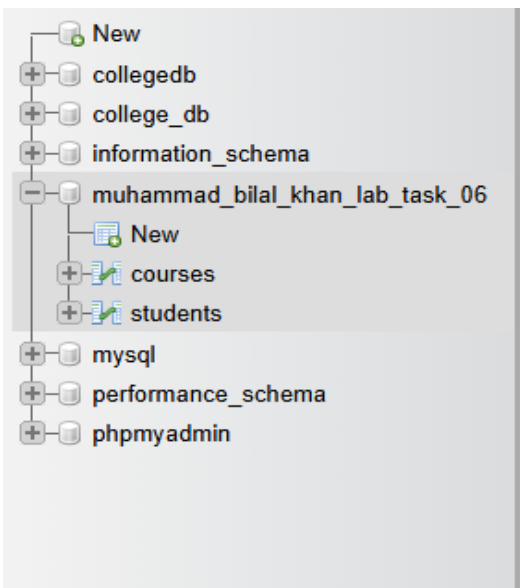
Alter the Second Table



✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0278 seconds.)

```
ALTER TABLE courses ADD CourseDescription TEXT;
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)



```
1 ALTER TABLE courses MODIFY Credits DECIMAL(3,1);
```

New

collegedb

college_db

information_schema

muhammad_bilal_khan_lab_task_06

New

courses

students

mysql

performance_schema

phpmyadmin

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0247 seconds.)

ALTER TABLE

courses

MODIFY

Credits

DECIMAL(3,1);

[Edit inline]

[Edit]

[Create PHP code]

New

collegedb

college_db

information_schema

muhammad_bilal_khan_lab_task_06

New

courses

students

mysql

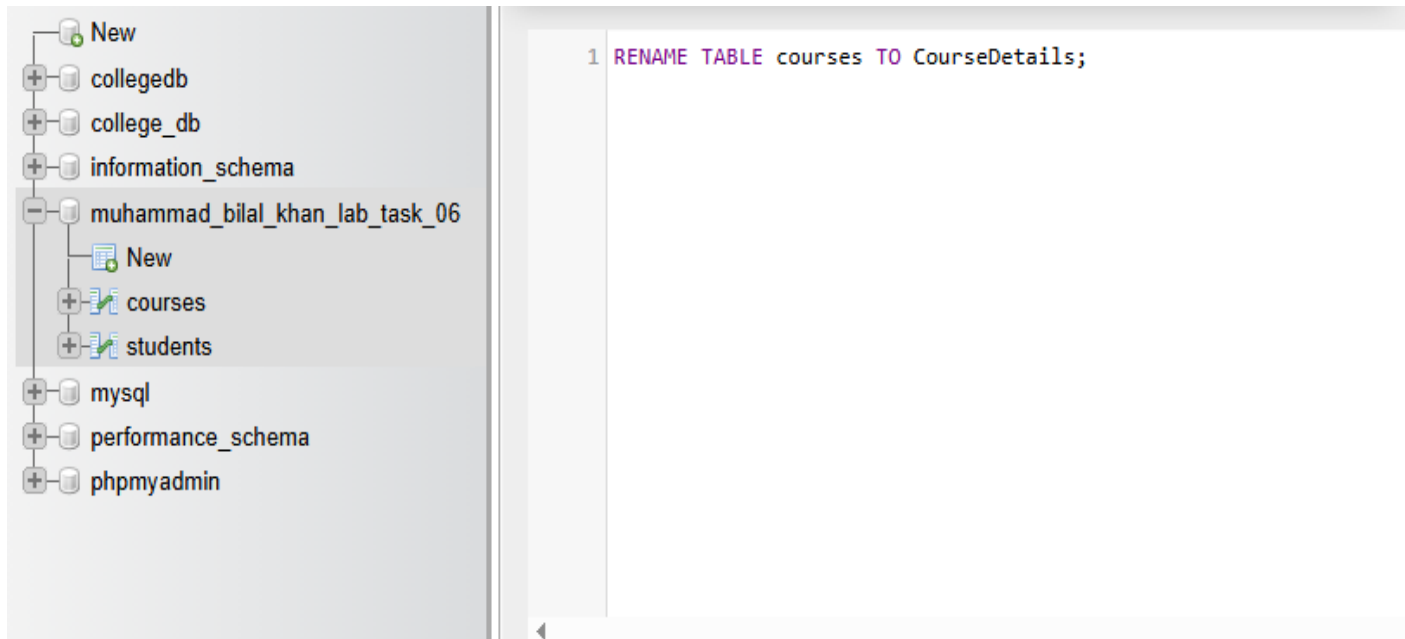
performance_schema

phpmyadmin

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	CourseID 	int(11)			No	None			 Change  Drop  More
☐ 2	CourseName	varchar(100)	utf8mb4_general_ci		No	None			 Change  Drop  More
☐ 3	Instructor	varchar(50)	utf8mb4_general_ci		No	None			 Change  Drop  More
☐ 4	Credits	decimal(3,1)			Yes	NULL			 Change  Drop  More
☐ 5	StartDate	date			Yes	NULL			 Change  Drop  More
☐ 6	EndDate	date			Yes	NULL			 Change  Drop  More
☐ 7	Department	varchar(50)	utf8mb4_general_ci		No	None			 Change  Drop  More
☐ 8	CourseDescription	text	utf8mb4_general_ci		Yes	NULL			 Change  Drop  More

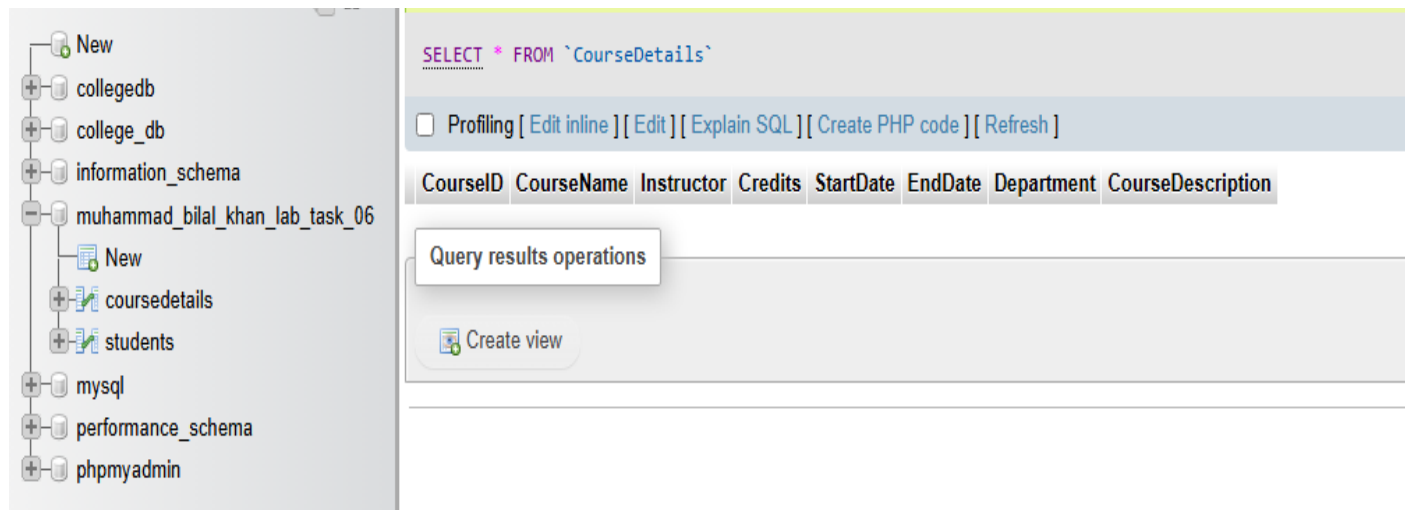
Section 5:

Rename and Delete Tables and Database



The screenshot shows the MySQL Workbench interface. On the left, the 'Schemas' pane displays a tree structure with databases: 'collegedb', 'college_db', 'information_schema', 'muhammad_bilal_khan_lab_task_06', 'mysql', 'performance_schema', and 'phpmyadmin'. Under 'muhammad_bilal_khan_lab_task_06', there is a 'New' folder containing 'courses' and 'students' tables. The main SQL editor on the right contains the following query:

```
1 RENAME TABLE courses TO CourseDetails;
```



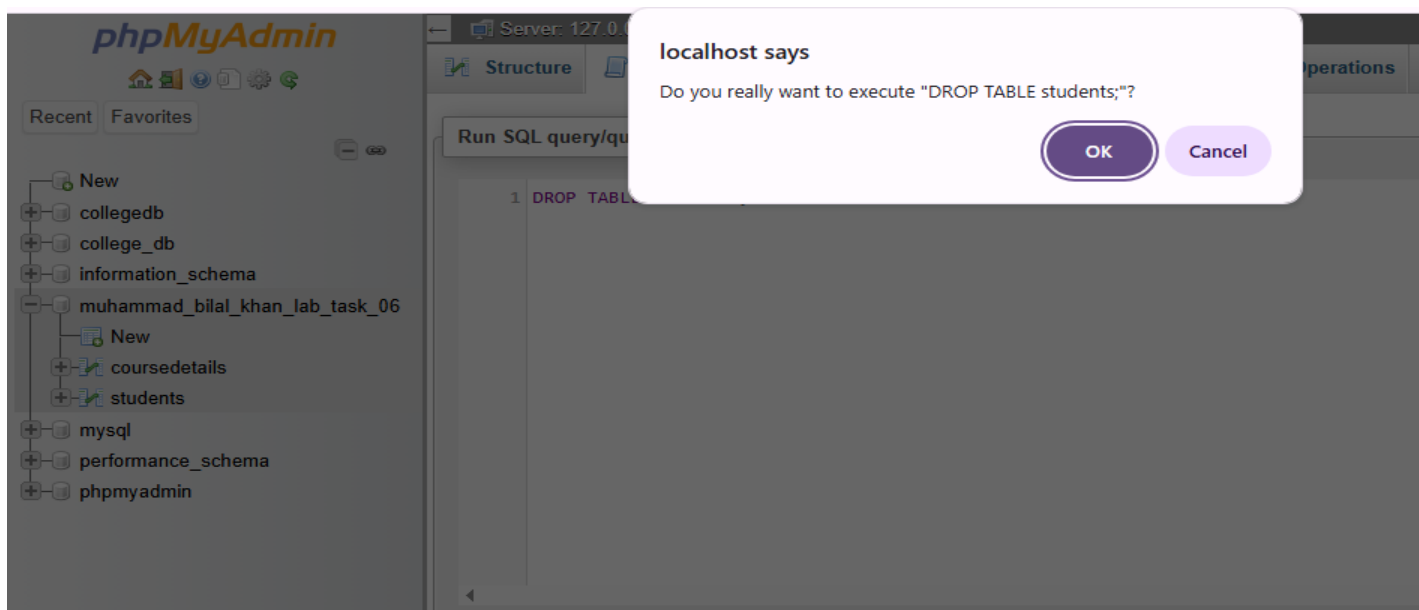
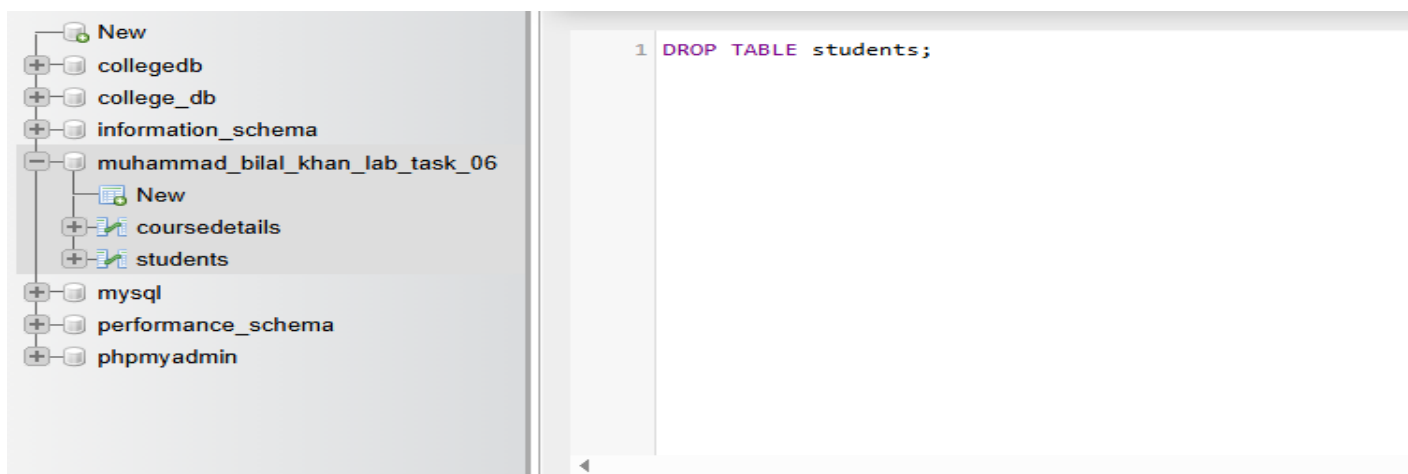
The screenshot shows the MySQL Workbench interface after executing the query. The 'Schemas' pane on the left is identical to the previous screenshot, but now the 'New' folder under 'muhammad_bilal_khan_lab_task_06' contains 'coursedetails' and 'students' tables. The main SQL editor on the right displays the query:

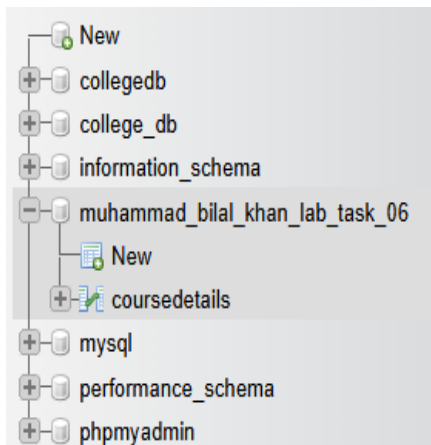
```
SELECT * FROM `CourseDetails`
```

Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. The query results are displayed in a table with the following columns:

CourseID	CourseName	Instructor	Credits	StartDate	EndDate	Department	CourseDescription
----------	------------	------------	---------	-----------	---------	------------	-------------------

Below the table, there is a 'Query results operations' section with a 'Create view' button.

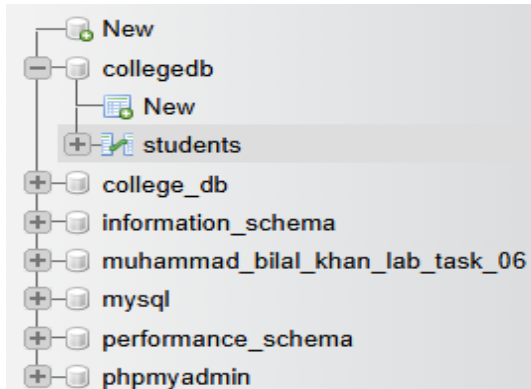




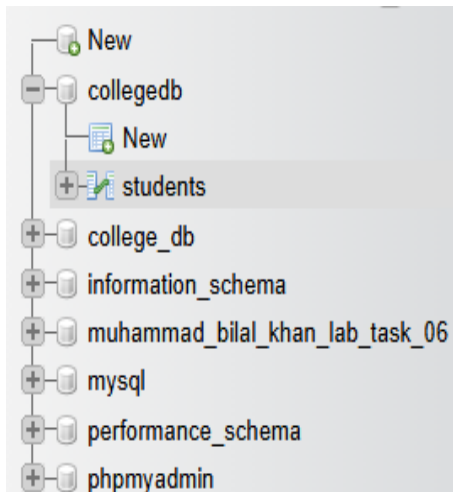
✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0276 seconds.)

```
DROP TABLE students;
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]



```
1 USE collegedb;  
2  
3 DROP DATABASE collegedb;
```



✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

```
USE collegedb;
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0316 seconds.)

```
DROP DATABASE collegedb;
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

New

+

college_db

+

information_schema

+

muhammad_bilal_khan_lab_task_06

+

mysql

+

performance_schema

+

phpmyadmin

Create database

Database name

utf8mb4_general_ci

Create

☐ Check all

Drop

Database	Collation	Action
☐ college_db	utf8mb4_general_ci	Check privileges
☐ information_schema	utf8_general_ci	Check privileges
☐ muhammad_bilal_khan_lab_task_06	utf8mb4_general_ci	Check privileges
☐ mysql	utf8mb4_general_ci	Check privileges
☐ performance_schema	utf8_general_ci	Check privileges
☐ phpmyadmin	utf8_bin	Check privileges

Total: 6